AI-as-Labor Operating Model: Cross-Domain Ideation

Status: Ideation / Exploration
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Purpose

This document defines the **AI-as-Labor operating model** and establishes a **strict, falsifiable framework** for evaluating whether that model can generalize beyond demand planning into other enterprise planning domains.

This document exists to:
* Test structural transferability of the operating model
* Identify domains that are ineligible by design
* Prevent premature or speculative expansion narratives

This document does not:
* Propose category expansion
* Propose product roadmap changes
* Establish execution priorities
* Signal expansion intent

This is constrained ideation, not strategy.

Hard Constraints from Daybreak Strategy

All cross-domain ideation must satisfy **every** constraint below. Failure on any single constraint **disqualifies the domain**.

1. Decision Ownership Must Be Explicit
A domain must support clearly scoped, repeatable decisions with a single accountable owner. If ownership is implicit, shared, or negotiated, the domain is ineligible.

2. Judgment Must Compound
Human overrides must be:
* Capturable with context
* Scorable against outcomes
* Absorbable or rejectable by the system
If judgment cannot improve future decisions, the system is automation, not labor.

3. Governance Precedes Autonomy
Autonomy must be enforced through systemic policy, not informal human review. Domains that rely on universal review, consensus workflows, or procedural safeguards violate this constraint.

Operating Model Failure Modes (Non-Negotiable)

AI-as-Labor is only valid if its failure modes are detectable, bounded, and enforceable.

Canonical Failure Modes
1. Silent Drift - Outcomes degrade while decisions remain autonomous.
2. Judgment Staleness - Human overrides no longer outperform the agent baseline.
3. Override Dependency - The same decision identity requires repeated intervention across cycles.
4. Policy Overfitting - Governance rules begin to approximate decision logic instead of risk control.
5. Trust Collapse - Humans bypass the system through shadow processes or offline edits.

Invalidation Rule
If a failure mode cannot be detected through system metrics (not audits or anecdotes), the domain is ineligible for AI-as-Labor.

Domain Eligibility Filter (Hard Gate)

A domain may only be considered for AI-as-Labor applicability if **all** conditions below are satisfied. The default assumption is that **most domains fail**.

1. Decision Atomicity Exists - Decisions can be owned at a bounded, repeatable unit of work.
2. Outcome Feedback Is Timely - Outcomes resolve within a governance-tolerable learning cycle.
3. Baseline Error Is Financially Bounded - Incorrect decisions are reversible without catastrophic consequences.
4. Consensus Is Not the Value Mechanism - Value is created through judgment application, not negotiation.

Passing this filter is necessary but **not sufficient** for expansion consideration.

What is the AI-as-Labor Operating Model?

AI-as-Labor is an operating model in which AI agents own **scoped, repeatable enterprise decisions** as the default decision-maker, while humans supervise through policy, risk thresholds, and exception handling.

Core consequences
* The agent is accountable for the baseline outcome

* The human is accountable for governance quality
* If turning off the Agent does **not** require reassigning decision ownership to a human role, it is **not** AI labor

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Decision Identity Specification

Judgment can only compound if decisions have stable identity across cycles.

Decision Identity Requirements

Every AI-owned decision must be uniquely identified by:
* Decision type
* Entity scope (e.g., SKU, location)
* Temporal horizon
* Version lineage
* Agent version hash

Identity Persistence Rules
* Re-aggregation or re-segmentation must preserve backward traceability
* Decisions without resolvable prior outcomes cannot learn
* Identity collisions invalidate override learning

What AI-as-Labor Is
AI qualifies as labor only if **all** conditions below are met.

1. Decision Ownership Is Explicit
* The agent is the named owner of the baseline decision
* Outcomes are attributable to the agent, not “the system”
* Humans are not implicitly accountable for machine output

2. AI Does the Default Work
* The agent processes 100% of decisions by default
* Humans intervene only on exceptions, or policy breaches

Inversion:
* Before: Humans work → AI assists
* After: AI works → Humans supervise
If humans must review everything, labor has not shifted.

3. Governance Is Native
Autonomy is:

* Scoped
* Auditable
* Reversible at the **decision** level

If safety depends on informal human review, the AI does not own the work.

Override Quality Grading

Not all overrides represent superior judgment.

Override Evaluation Dimensions
* Intent Validity: Was the rationale coherent at decision time?
* Information Superiority: Did the human introduce unavailable information?
* Outcome Delta: Did the override outperform the agent baseline?
* Consistency: Are similar contexts handled similarly?

Learning Rule
* Lucky overrides do not train the system
* Harmful overrides are penalized regardless of confidence

** Accountability Boundary**

AI Owns
* Baseline execution
* Routine decision output

Humans Own
* Policy design quality
* Risk threshold calibration
* Override clarity and timeliness

Humans are ***not*** accountable for baseline decision accuracy or downstream feasibility outcomes.

What AI-as-Labor Is Not

AI-as-Labor is **not**:
* A copilot
* A recommendation engine
* An optimization module
* A forecasting tool
* A workflow accelerator

Those improve software. AI-as-Labor replaces execution capacity while preserving control.

Domain Anti-Patterns (Explicitly Ineligible)

The following domains often appear eligible but fail by design:
* Consensus-driven planning
* Safety-critical control systems
* Strategic one-off decisions
* Long-latency outcome environments

If value depends on alignment rather than decision accuracy, the domain is not AI-as-Labor.

Demand Planning as the Wedge
Demand planning is the **proof point**, not the endpoint.
Judgment breaks here first:
* Highest override rates
* Most manual effort
* Largest financial impact
If judgment cannot scale here, it cannot scale anywhere.

Forecast Baseline Pack (FBP) (Mathematical Grounding)

Definition:
Forecast Baseline Pack (FBP) = a structured, versioned set of model-generated forecast candidates plus diagnostics produced **before** agent reasoning.

Role in the operating model:
FBP is **non-authoritative** and cannot be published as the Unconstrained Forecast (UCF). It is grounding evidence the agent may accept, reject, or blend. Its presence enables falsifiable learning (overrides scored against outcomes relative to baseline candidates).

Governance-safe rule:
Every published UCF must reference the FBP hash used at decision time to preserve decision identity and enable outcome attribution.

FBP contents (minimum)

1) Baseline forecast candidates (one or more)
* `candidate_id`
* `yhat[sku, location, time_bucket]` at the UCF decision granularity
* Optional: `yhat_components` (trend / seasonality / promo / price attribution) for explainability

2) Uncertainty and distribution
* `quantiles[p10, p50, p90]` or `prediction_interval_low/high`
* Optional: `probability_of_change` or `regime_shift_score`

3) Backtest and reliability diagnostics
* `error_metrics` by horizon and hierarchy (MAPE, WAPE, bias)
* `calibration_metrics` (interval coverage)
* `recent_drift_metrics` (rolling bias, rolling error)

4) Data quality diagnostics
* missingness, outlier flags, late-arriving actuals, promo-calendar integrity
* “signal integrity” flags the agent should treat as grounding defects, not demand shifts

5) Versioning and reproducibility
* `model_version_hash`, `feature_set_hash`, `training_cutoff_date`, `data_snapshot_hash`

Demand Planning Decision Definition: Unconstrained Forecast (UCF)

AI-Owned Decision

For the Demand Planning use case, the AI agent owns the **Unconstrained Forecast (UCF)**.

Forecast Baseline Pack (FBP) = a structured, versioned set of baseline forecast candidates (model-generated and optionally business-provided) plus diagnostics produced before agent reasoning; the agent-owned UCF is not a model output and is not constrained by baseline candidates.

The Unconstrained Forecast is:
* The best mathematical + judgment-informed estimate of true market demand
* Independent of supply, capacity, inventory, financial targets, or execution feasibility
* Produced prior to any cross-functional reconciliation or constraint application

This decision is owned end-to-end by the agent within its scoped horizon and granularity.

Explicit Ownership Boundary

The Demand Planning Agent does **not** own:
* Supply-constrained forecasts
* Feasible or executable plans
* Inventory, production, or capacity allocation
* Financial targets or commitments
* S&OP or IBP consensus outputs

Governance Model for the UCF

Human supervision of the UCF is permitted only to:
* Correct demand signal interpretation
* Inject event-based judgment outside the Agents knowledge
* Enforce policy and risk thresholds

Human intervention may **not** introduce supply, capacity, inventory, or financial constraints

System Configuration
* **Planning Cycle**: Time window in which decisions are evaluated against policy
* **Grounding**: Master, transactional, and planning data integration, including:
  * Historical actuals and order signals
  * Known events (promo calendar, launches, discontinuations)
  * External drivers (price, macro, weather where relevant)
  * **Forecast Baseline Pack (FBP)** (see section above)
* **Scope**: Decisions assigned to a user
* **Policy**: Company-defined exception conditions and risk thresholds
* **Playbook**: User-provided contextual documents
* **Context Parsing**: Natural-language summarization for system prompt
* **User Context**: User-provided natural-language input
* **System Prompt**: Combined grounding and contextual inputs for agent reasoning

Supervision Loop (Not an Approval Workflow)
1. **Observe**: Review suggested decisions outside policy, prioritized by impact
2. **Orient**: Understand agent decision, reasoning, and actions
   * Compare agent UCF vs FBP candidate(s)
   * Inspect uncertainty bands and drift diagnostics
   * Inspect component attribution for large deltas (promo/price/seasonality/trend)
   * Confirm whether anomalies are demand shifts or data quality defects
3. **Decide**: Accept or override with reason, confidence, and explanation
4. **Act**: Commit the decision

This loop describes **supervision mechanics**, not approval workflows.

Decision Status Paths
Suggested → Reviewed → Accepted → Submitted
Suggested → Reviewed → Overridden → Submitted
Autonomous → Reviewed → Accepted → Submitted
Autonomous → Reviewed → Overridden → Submitted
Autonomous → Not Reviewed → Auto-Submitted

Policy Logic

Suggested =

(Policy check(s) fails AND $ impact > risk threshold)
OR (AI decision delta > risk threshold)

Demand Planning Agent Policies

Policies are formal governance constraints that determine whether an AI-owned Unconstrained Forecast (UCF) may be auto-submitted or must be surfaced for human supervision.

Policies do not modify forecasts. They only control autonomy and Decision Status Path.

Policy Class: Demand Signal Integrity

These policies ensure the Unconstrained Forecast remains a credible representation of true market demand before any downstream constraints are applied.

Violations do not invalidate the forecast. They may trigger supervision when risk thresholds are exceeded.

Policy	Purpose	Trigger Condition	Governance Action
Forecast Stability Policy	Prevents excessive volatility inconsistent with demand behavior	Period-over-period change exceeds configured tolerance	Mark decision as **Suggested**
History Anchoring Policy	Ensures forecast does not drift materially from recent demand reality	Deviation from rolling actuals exceeds threshold	Mark decision as **Suggested**
Trend Consistency Policy	Guards against unexplained trend reversals	Slope change exceeds tolerance vs historical trajectory	Mark decision as **Suggested**
Seasonality Alignment Policy	Prevents loss of known seasonal structure	Correlation to historical seasonality below threshold	Mark decision as **Suggested**

Risk Threshold Framework

Risk thresholds determine whether a policy violation requires human supervision based on potential business impact.

Risk thresholds:
* Do not Modify the Unconstrained Forecast
* Do not encode feasibility logic
* Govern escalation priority only

Risk Dimensions

Risk is evaluated along one or more of the following dimensions:

* **Financial Exposure**: Expected dollar impact if the forecast is wrong
* **Volume Exposure**: Absolute unit deviation at risk
* **Change Magnitude**: Size of deviation from prior baseline
* **Volatility Sensitivity**: Historical instability of the item or segment
* **Business Criticality**: Strategic importance (e.g., launch items)

Thresholds may be combined but must remain interpretable.

Enforcement Rules

* Policy violations below risk thresholds remain **Autonomous**
* Policy violations above risk thresholds are marked **Suggested**
* Multiple low-risk violations increase priority, not authority

Escalation allocates attention, not decision ownership.

Key Metrics

* **Touchless Rate**: % of decisions not reviewed
* **Agent Decision Quality Score (Agent DQS)**: Outcome lift from agent decisions
* **User Decision Quality Score (User DQS)**: Outcome lift from human overrides
* **Cognitive Load Index (CLI)**: Decisions reviewed divided by total decisions in scope
* **Override Dependency Ratio (Kill Metric)**: Repeated human intervention on the same decision identity

If value depends on persistent human intervention, the system is **not** AI-as-Labor.

Allocation Planning as the next AI Labor opportunity
Allocation planning is eligible for AI-as-Labor only when allocation is treated as a **scoped, repeatable decision** with explicit ownership and systemic governance.
* The agent owns the baseline allocation decision by default.
* Humans supervise through policy and risk thresholds, not universal review.
* Overrides are captured, scored against outcomes, and either absorbed or rejected by the system.

Solver Baseline Pack (SBP) (Mathematical Grounding)

Definition:
Solver Baseline Pack (SBP) = a structured, versioned set of solver-generated **candidate Allocation Commits** plus **constraint and value diagnostics** produced **before** agent reasoning.

Role in the operating model:
SBP is **non-authoritative** and cannot be published to execution as the Allocation Commit (AC). It is grounding evidence the Allocation Planning Agent may accept, reject, or blend when producing the agent-owned AC. This preserves explicit decision ownership and makes overrides scorable against outcomes relative to the solver baselines.

SBP contents

1) Candidate Allocation Commits (one or more)
Each candidate is a complete AC at the decision granularity.

* `candidate_id` (e.g., `FAIR_SHARE_V1`, `EQUAL_DOS_V1`, `PRIORITY_HEURISTIC_V2`, `LP_V1`, `MILP_V1`, `HYBRID_V2`)
* `entitlement_qty [item, segment, time_bucket]`
* `allocation_priority [item, segment, time_bucket]`
* `unallocated_qty [item, time_bucket]`
* Optional but preferred: `pegging_map [supply_pool -> demand_segment]` and/or `pegging_summary`
* Optional: `split_count_stats` (to detect pathological fragmentation)

2) Feasibility and constraint diagnostics (machine-readable)
These are required so the agent can explain “why shorted” and detect unsafe outputs.

* `feasibility_status` = `PASS | FAIL`
* `infeasibility_reasons[ ]` (if `FAIL`)
* `binding_constraints[ ]` (material, capacity, lead time, eligibility, shelf-life, substitution)
* `slack_by_constraint[ ]` (where slack exists)
* `gating_factors[item, time_bucket]` (dominant constraint causing shortfall)
* Optional: `constraint_violations[ ]` (should be empty for valid candidates)

3) Marginal value signals (for tradeoff reasoning)
Allocation equivalent of forecast uncertainty: “where is 1 extra unit worth the most.”

* `shadow_price_supply[item, time_bucket]` (LP duals or calibrated approximation)
* `shadow_price_capacity[resource, time_bucket]`
* `marginal_short_penalty[item, segment, time_bucket]` (implied cost of shorting)
* Optional: `substitution_premium[item, alt_item]`

4) Policy compliance metrics computed on each candidate
Policies still do not change AC values; these metrics inform autonomy and review prioritization.

* `service_floor_breaches[]` (magnitude + exposure)
* `fair_share_deviation[]`

* `priority_inversions[]`
* `volatility_delta_vs_prior[]` (entitlement delta magnitude)

5) Versioning and reproducibility
SBP must be reproducible for audit and learning.

* `solver_version_hash`
* `objective_config_hash` (weights/penalties used in candidate solves)
* `constraint_set_hash`
* `input_snapshot_hash` (supply, demand, master data)
* `solve_timestamp`
* Optional: `runtime_stats` (solve time, gap, iterations)

Governance-safe rule:
SBP candidates are mathematical baselines used for grounding and comparison. The agent-owned Allocation Commit is the only publishable decision artifact, and every published AC must reference the SBP hash used at decision time to preserve decision identity and enable outcome attribution.

Allocation Planning Decision Definition: Allocation Commit (AC)

AI-Owned Decision
For the Allocation Planning use case, the AI agent owns the **Allocation Commit (AC)**.

The Allocation Commit is defined as:
* The best **constraint-respecting** distribution of constrained finished goods supply across demand segments and time buckets
* Expressed as **entitlement caps** and **priority ordering** that downstream execution must enforce
* Produced **prior to** order-level promising, backlog negotiation, or cross-functional consensus

Minimum decision artifact:
* `entitlement_qty [item, segment, time_bucket]`
* `allocation_priority [item, segment, time_bucket]`
* `unallocated_qty [item, time_bucket]`
* `pegging_summary [supply pool to segment attribution and gating factors]`

Explicit Ownership Boundary
The Allocation Planning Agent does **not** own:
* Segmentation definitions, customer tiering, contractual entitlement policy, fairness definitions, or strategic account doctrine
* Creating supply (expedites, overtime, re-sourcing), changing capacity, or changing lead times as a decision outcome
* Demand shaping decisions (pricing, promo cancellation, channel strategy)

* S&OE/S&OP reconciliation outputs and stakeholder alignment artifacts
* Order-level promising and backlog sequencing decisions (owned by a separate execution agent or system)

The agent **does** own:
* Applying the approved policy bundle and constraints to produce the Allocation Commit
* Explaining allocations through pegging and gating-factor attribution
* Detecting when the decision is unsafe to auto-submit under policy

Governance Implication
Human supervision of the Allocation Commit is permitted only to:
* Correct missing or wrong grounding inputs (inventory, supply dates, lead times, eligibility, shelf-life, substitution permissions)
* Inject documented exception context not available to the agent (contract penalties, recalls, regulatory holds, CEO-level commitments) with explicit reason and evidence
* Tune policy and risk thresholds

Human intervention may **not**:
* Convert allocation into a consensus workflow where negotiation is the mechanism of value creation
* Introduce new segmentation or entitlement policy ad hoc through manual edits
* Repeatedly patch the same decision identity without upstream policy or data correction (this is an operating model failure signal)

System Configuration
* **Planning Cycle**: Time window in which Allocation Commits are evaluated against policy and later scored against outcomes
* **Grounding**:
  * Inventory and supply (on-hand/in-transit/production)
  * Lead times, capacity limits, shelf-life, regulatory eligibility, substitution rules
  * Demand by segment (orders, backlog, forecast consumption where applicable)
  * Policy bundle (entitlements, priorities, floors/ceilings, fairness rules)
  * **Solver Baseline Pack (SBP)**: mathematical grounding produced prior to agent reasoning
    * **Candidate Allocation Commits** (heuristic / optimization / hybrid outputs)
    * **Constraint Diagnostics** (binding constraints, slacks, gating factors, feasibility certificate)
    * **Marginal Value Signals** (shadow prices / marginal penalties for supply and capacity)
    * **Policy Compliance Metrics** (floor breaches, fairness deviation, priority inversions, volatility deltas)
* **Scope**: Items, locations, supply pools, segments, and horizons assigned to a user or team
* **Policy**: Company-defined exception conditions and risk thresholds that determine autonomy
* **Playbook**: Contracts, allocation doctrine, service floors, escalation rules, and exception handling guidance

* **Context Parsing**: Summarization of playbooks and recent overrides into decision-time context
* **User Context**: User-provided notes about current cycle anomalies (strikes, port closures, promos, recall holds)
* **System Prompt**: Combined grounding and contextual inputs for agent reasoning

Supervision Loop (Not an Approval Workflow)
1. **Observe**: Review decisions outside policy, prioritized by business impact
2. **Orient**:
   * Compare final Allocation Commit vs SBP candidate(s)
   * Inspect gating factors, binding constraints, and pegging explanations
   * Review marginal value signals (where 1 unit of supply/capacity is worth the most)
3. **Decide**: Accept or override with reason, confidence, and evidence
4. **Act**: Commit and publish the Allocation Commit for execution

Review is exception-driven, not mandatory.

Decision Status Paths
Suggested → Reviewed → Accepted → Submitted
Suggested → Reviewed → Overridden → Submitted
Autonomous → Reviewed → Accepted → Submitted
Autonomous → Reviewed → Overridden → Submitted
Autonomous → Not Reviewed → Auto-Submitted

Policy Logic
Suggested =
(Policy check(s) fails AND $ impact > risk threshold)
OR (Allocation Commit delta > risk threshold)
OR (Allocation integrity violation is detected)

Allocation Planning Agent Policies
Policies are formal governance constraints that determine whether an AI-owned Allocation Commit may be auto-submitted or must be surfaced for human supervision.

Policies do not modify allocations. They only control autonomy and review priority.

Policy Class: Allocation Integrity
These policies ensure the Allocation Commit is internally coherent, feasible under modeled constraints, and explainable through pegging.

Policy	Purpose	Trigger Condition	Governance Action

Supply Conservation Policy	Prevents allocating more than available supply	Allocated qty exceeds available supply in any pool/bucket	Mark decision as **Suggested** and **Block submission**
Temporal Feasibility Policy	Prevents impossible availability timing	Allocation implies ship/consume before availability or lead-time feasibility	Mark decision as **Suggested** and **Block submission**
Eligibility & Compliance Policy	Prevents prohibited allocations	Allocation violates regulatory/market eligibility constraints	Mark decision as **Suggested** and **Block submission**
Pegging Explainability Policy	Ensures shortfalls are attributable	Unallocated demand cannot be attributed to a constraint or policy rule	Mark decision as **Suggested**

Policy Class: Allocation Outcome Risk
These policies govern whether the Allocation Commit is safe to auto-submit given business exposure.

Policy	Purpose	Trigger Condition	Governance Action
Service Floor Compliance Policy	Enforces minimum service commitments by segment	Any segment floor is breached beyond tolerance	Mark decision as **Suggested**
Fair-Share Deviation Policy	Prevents unfair distribution beyond policy	Deviation from fair-share target exceeds tolerance	Mark decision as **Suggested**
Priority Inversion Policy	Prevents non-policy priority reversals	Lower-priority segment receives supply while higher-priority segment is shorted beyond tolerance	Mark decision as **Suggested**
Allocation Stability Policy	Prevents unexplained whiplash	Period-over-period entitlement change exceeds tolerance	Mark decision as **Suggested**
Expiry / Obsolescence Exposure Policy	Prevents avoidable write-offs	Allocation increases projected expiry/write-off beyond threshold	Mark decision as **Suggested**

Policy Enforcement Rules
* Policy violations never change the Allocation Commit.
* Policy violations only affect review status (Autonomous vs Suggested) and, for integrity violations, submission eligibility.
* Multiple policy violations increase review priority, not authority.
* No policy may introduce new segmentation, entitlement logic, or stakeholder-specific negotiation rules.

Policy Validity Test
A rule qualifies as an Allocation Planning Agent policy only if:
* Removing the rule does not change the Allocation Commit values
* The rule can be enforced without cross-functional negotiation
* The rule applies uniformly across items/segments within scope

* The rule preserves the Allocation Commit as an execution of approved policy plus constraints, not a proxy consensus workflow

Risk Threshold Framework
Risk thresholds determine whether a policy violation requires human supervision based on potential business impact.

Risk thresholds do not:
* Modify allocations
* Validate correctness
They only govern escalation and prioritization.

Risk Dimensions
Risk is evaluated along one or more of the following dimensions:
* **Revenue Exposure**: Expected revenue at risk if the allocation is wrong
* **Service Exposure**: Expected OTIF or fill-rate degradation on critical segments
* **Penalty Exposure**: Contractual penalties, chargebacks, SLA breaches
* **Churn Proxy**: Customer risk weighting for strategic accounts
* **Expedite Cost Exposure**: Premium freight, overtime, substitution premium
* **Compliance Exposure**: Regulatory or recall-related risk

Thresholds may be combined but must remain interpretable.

Enforcement Rules
* Policy violations below risk thresholds remain **Autonomous**
* Policy violations above risk thresholds are marked **Suggested**
* Multiple low-risk violations do not automatically escalate severity
* Risk thresholds affect priority, not authority

Explicit Exclusions
Risk thresholds may not:
* Encode stakeholder negotiation outcomes
* Represent financial targets or commitments as decision logic
* Act as a proxy approval workflow
* Select winners and losers by embedding hidden allocation logic

Risk Threshold Validity Test
If adjusting a risk threshold changes the Allocation Commit values, the threshold is mis-specified and invalid.

Key Metrics
* **Touchless Rate**: % of Allocation Commits not reviewed
* **Agent Decision Quality Score (Agent DQS)**: Outcome lift attributable to agent Allocation Commits

* **User Decision Quality Score (User DQS)**: Outcome lift from human overrides vs the agent baseline
* **Cognitive Load Index (CLI)**: Decisions reviewed divided by total decisions in scope
* **Override Dependency Ratio (Kill Metric)**: Repeated human intervention on the same decision identity across cycles
* **Execution Adherence**: Divergence between committed entitlements and executed fulfillment (proxy for trust collapse)